

LOW-COST RESCUE DRONE

Engineering Project Overview | For academic discussion with Professor Ian Piumarta, KUAS



Period July 2024 - January 2025	Team 2 members; I was project lead
Lab Atal Tinkering Lab, Aristo Public School	Final controller KK2.1.5
Focus Electronics, embedded systems, UAV control	Outcome Working prototype + flight demonstration

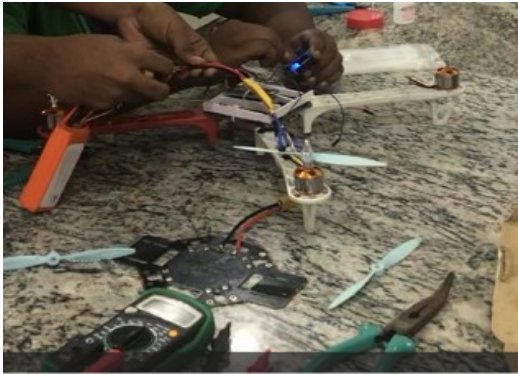
Completed prototype during electronics integration and assembly.

Project in brief

I began this project to understand how a quadcopter works by building and testing one myself. The project progressed from an ESP32 + MPU6050 concept, to an Arduino + MPU6050 experimental stage, and finally to a KK2.1.5-based working prototype. I led the technical development, hardware integration, assembly, testing, troubleshooting and documentation.

Prototype evolution

1	ESP32 + MPU6050	Embedded control, sensing and wireless-control exploration
2	Arduino + MPU6050	Stabilization, calibration, ESC control and radio integration
3	KK2.1.5	System integration, configuration, calibration and testing



Hands-on electronics work and testing.



Assembly and hardware integration.

Engineering work and learning

What I worked on

- Arduino-based experimental control and MPU6050 sensor integration
- Sensor calibration, ESC control and radio integration
- Wiring and integration of receiver, controller, ESCs, motors and power system
- KK2.1.5 configuration, calibration, testing and troubleshooting
- Project planning, documentation, teamwork and presentation

What I learned

- Engineering is iterative; the first approach does not always need to be the final one.
- Breaking a large system into smaller subsystems helps troubleshooting.
- Reliable hardware integration depends on careful wiring, calibration and repeated testing.
- Building with real components increased my interest in microcontrollers, electronics, embedded systems and control.



Successful flight demonstration of the completed quadcopter prototype.

Technical scope and current limits

This was an educational working prototype, not a commercial rescue system. Autonomous navigation, AI-based detection, obstacle avoidance and thermal imaging were not implemented. Formal performance measurements such as flight time, altitude, range and payload were also not recorded.

Recognition and evidence

The project was demonstrated at school and presented to students. The portfolio also records recognition from school leadership and participation in Innovators Day 2024, a national-level project contest organized by Sri Manakula Vinayagar Engineering College.

Why this project led me to electronics

The project introduced me to microcontrollers, sensors, flight-control systems, ESCs, motors, radio communication and hardware integration. I now want to strengthen these foundations through undergraduate engineering study and continue developing practical electronics and embedded-systems skills. My longer-term interest is to apply engineering to healthcare technology.

Leandro P | Prospective Undergraduate Applicant - September 2027 | India